ML Model Deployment Strategies and Model Serving using Fast API

CLA 4 Assignment — Observation Report

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Introduction

In this assignment, a machine learning classification model was developed using the Titanic training dataset from Kaggle. The dataset contains passenger information such as passenger class, sex, age, number of siblings or spouses aboard, number of parents or children aboard, fare, and port of embarkation. The model uses these details to predict whether a passenger survived. The work also demonstrates different ways to use a trained model and how to make predictions available through a FastAPI service.

Observations

1. **How does Online Inference differ from Batch Inference?**
   Online inference is used when a prediction is needed for one record at a time. For example, passenger details can be provided to the model and the model returns a survival prediction for that passenger. Batch inference is used when several records are available together. In this assignment, 20 unseen passenger records are passed to the model in one operation, and the predictions are saved in batch_predictions.csv. Online inference is useful for an individual request, while batch inference is convenient for processing a group of records efficiently

2. **What is the difference between Streaming Inference and Batch Inference**?
   Streaming inference processes records one at a time as they arrive. In this assignment, 10 unseen passenger records are simulated as a stream, and a one-second delay is added between predictions. This makes each prediction appear separately. Batch inference, on the other hand, collects a group of records and processes them together. Therefore, streaming focuses on sequential processing of incoming records, whereas batch inference focuses on processing a fixed group of records in one run.

3. **Why is FastAPI used for model serving?**
   FastAPI is used to make the trained model accessible through HTTP API endpoints. The /health endpoint checks whether the service is running, and the /predict endpoint accepts passenger details and returns the model's prediction. Pydantic is used to validate the incoming values before they are passed to the model. For example, a valid passenger class is 1, 2, or 3. An invalid value such as 5 is rejected with a validation error. This allows another program to request predictions without needing to run the model code directly.

4. **Why should the same trained model be reused for all inference strategies?**

   The same saved model, model.pkl, should be used for online, batch, streaming, and API inference so that all predictions are produced by the same trained pipeline. Training separate models for each method could lead to differences because of different training data, preprocessing, or model settings. Reusing the saved pipeline also keeps the preprocessing steps consistent and makes the results easier to compare and maintain.

Conclusion

This assignment demonstrates the main stages of deploying a machine learning model. The Titanic dataset was used to train and evaluate a classification pipeline, which was saved for reuse. Predictions were generated using online, batch, and streaming inference. The model was then made available through FastAPI, with a health check endpoint, a prediction endpoint, and input validation. Together, these tasks show how a trained model can be reused in different inference scenarios and accessed through a REST API.